

DeepLearning.AI

Observing & Benchmarking Agents

Module 2

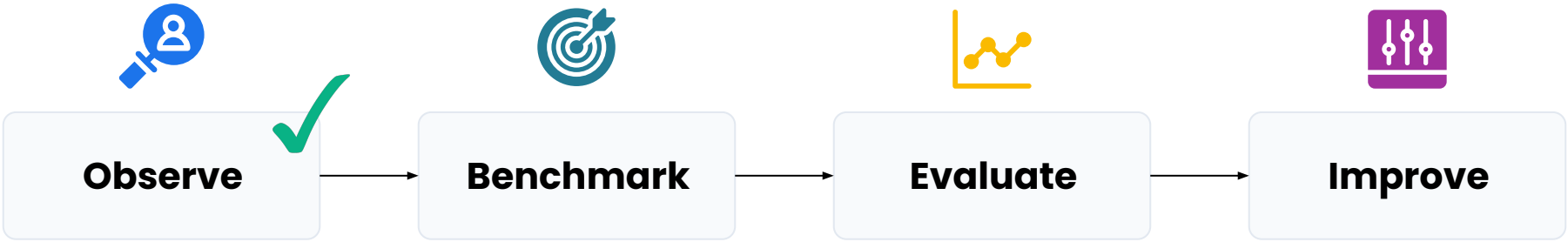


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Define application requirements

Module 2 · Video 1

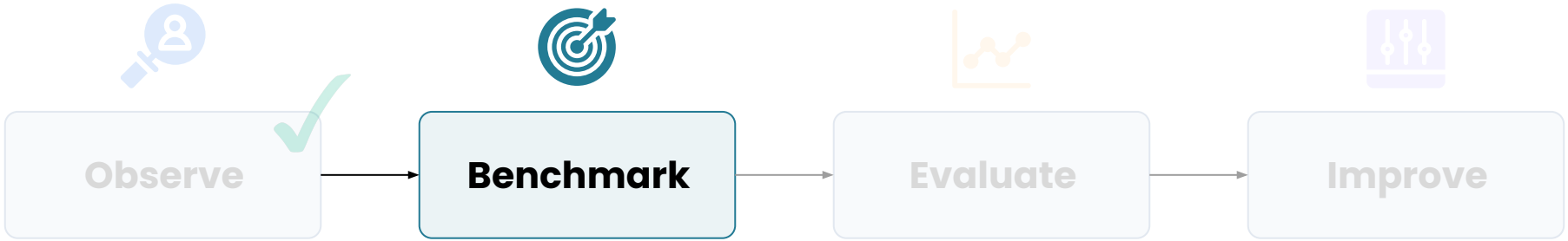
Compare actual vs. expected behavior



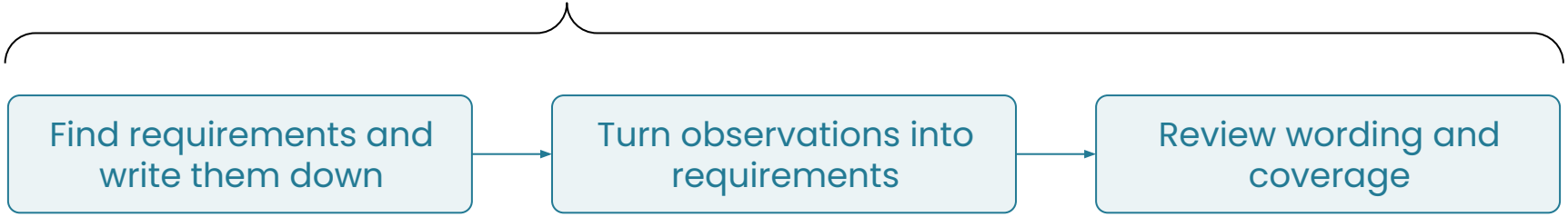
Capture what the agent did

Mistakes in agentic applications can be subtle

Find and develop requirements for your agent



Define what the agent should do and not do



Look for requirements in three places

Do-not-adopt flag



Reject



Which matches would you recommend and why?


```
{
 "volunteer_id": "2563",
 "dog_id": "182",
 "time": "SAT 10:00",
 "status": "confirmed"
}
```


Existing rules

People who know the work

Traces

Requirements

What should the application do or avoid doing?

A trace shows what the answer was based on

How should I care for my merino wool sweater?

Catalog:
`Machine washable on wool cycle`

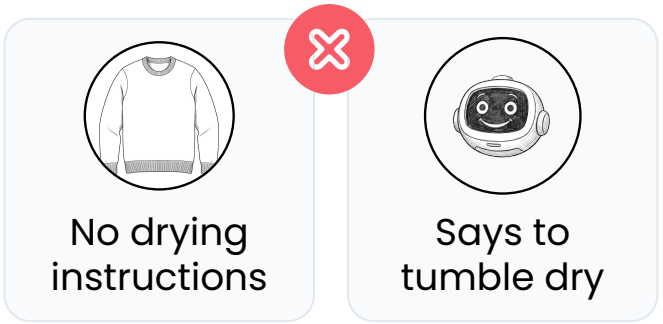


You can machine wash it on a wool cycle and then **tumble dry**



The agent recommended tumble drying, but the sweater didn't have drying instructions.

Turn one observation into a broader requirement



Don’t recommend tumble-drying this sweater

Too narrow

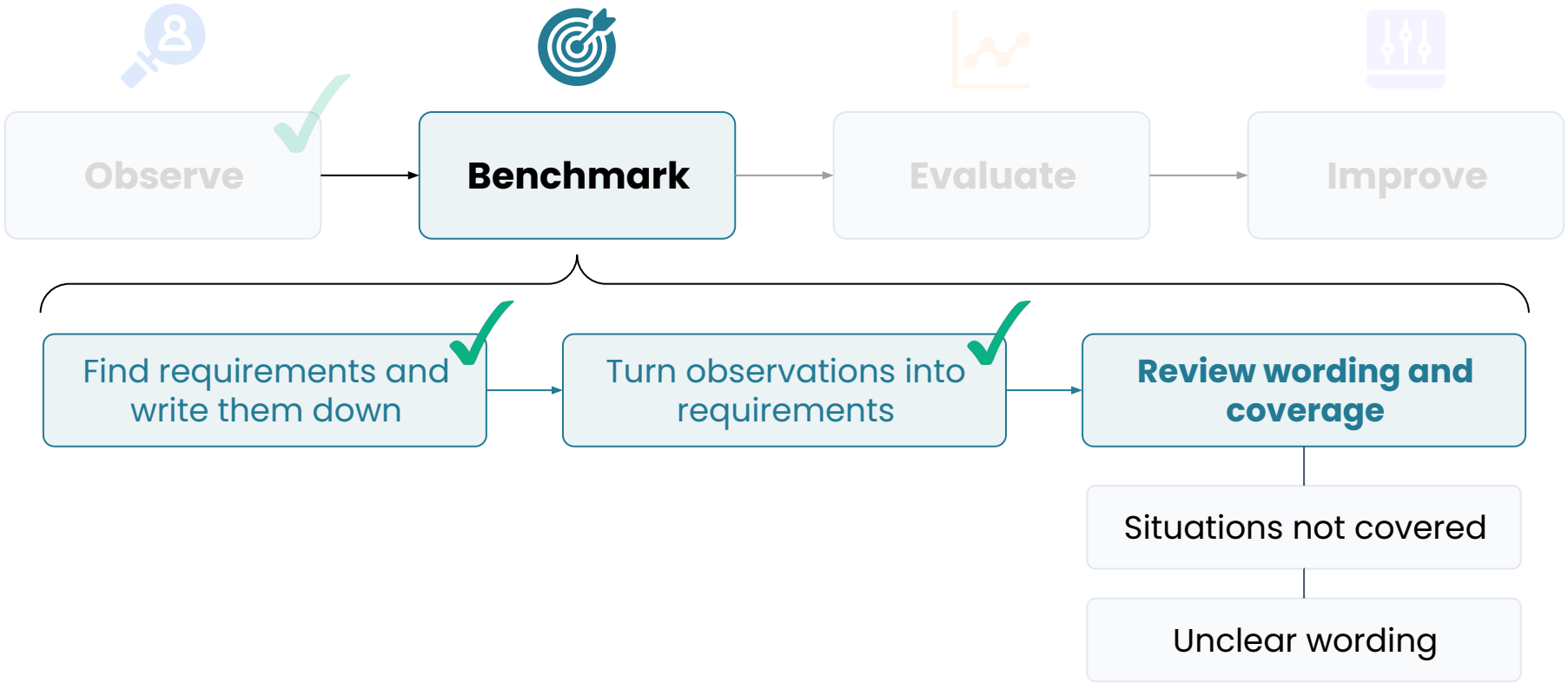
Only make claims about products that are supported by Northwind data

Broader requirement

Does the Northwind data support what the application told the customer?

Review question

Review and refine what you've written



Make vague words explicit

Vague requirement

Recommend a **suitable** dog

What makes a match unsuitable?



First-time handler



Experienced handlers only

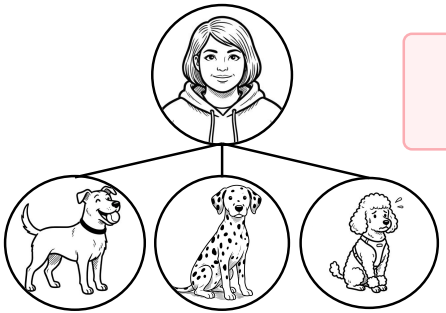
Explicit requirement

Recommend a suitable dog: **never match a dog that requires an experienced handler with a first-time volunteer**

Cover situations your requirements miss

Covered — Complete information

Not yet covered — Missing or conflicting information



No suitable dog available

Shelter rule?

- Yes: Add it to the requirements
- No: Decide what should happen

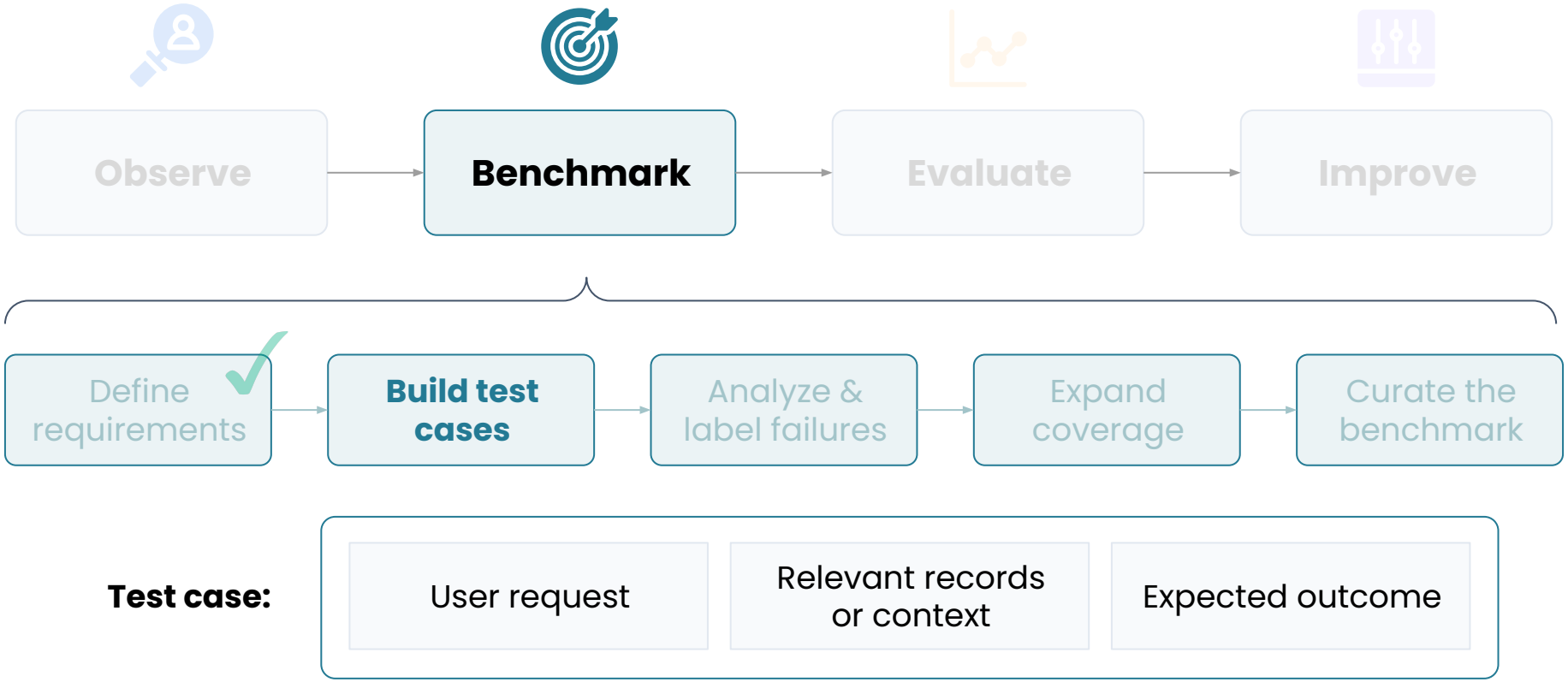


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Turn requirements into test cases

Module 2 · Video 2

Turn requirements into test cases



One requirement applies in many situations

Requirement

Only match volunteers with dogs they can safely handle

Experience levels



First time



Experienced

Handling needs



Calm



Experienced handlers only

Missing information



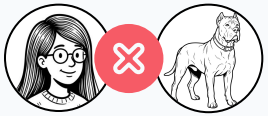
Experience not recorded



No handling notes

Test both: **Experience** + **Physical ability**

Find situations from different angles

Safe or unsafe match	What the application knows	How the request is worded
 First-time	**Handling notes:** experienced only	Can I take Buddy out for a walk?
 Experienced, physically able	**Handling notes:** [missing]	I'd like to take Buddy. Would that be okay?
 Experienced, dog too strong	**Note A:** beginner OK **Note B:** experienced only	can i wlk budy

Start with: likely situations, mistakes you've seen, risky cases

Each case pairs the facts with an expected outcome

User request	Volunteer & dog info	Expected outcome
Can I take Buddy out for a walk?	 Experienced  130 lb, too strong	**Reject** the match
Can I take Daisy out for a walk?	 Experienced  45 lb, calm	**Approve** the match, if eligible and available

Test cases can reveal requirement gaps

Requirement

Only match volunteers with dogs they can safely handle.



Experienced



No handling notes

Ask for more information?

or

Reject the match?

Check the shelter's policy

Or ask shelter staff



Once decided: add it to the requirement and the expected outcome

Not every case has one exact answer



Experienced. Can safely handle large dogs



45 lb, calm



Energetic, 100 lb



Reactive, experienced handler required

All three are acceptable matches

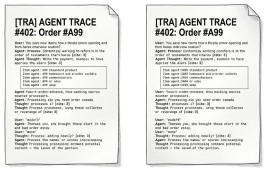
Suitable dog: the volunteer has the experience and physical ability to handle it

Example of one suitable response



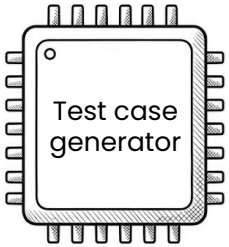
Daisy would be a good match: she's calm, and you've handled large dogs

Know where each example came from



Examples you already have

Source: real request



New examples

- ✓ Realistic
- ✓ Fit the situation

Source: LLM

Later, the source helps decide: weight of the case, in the benchmark or not

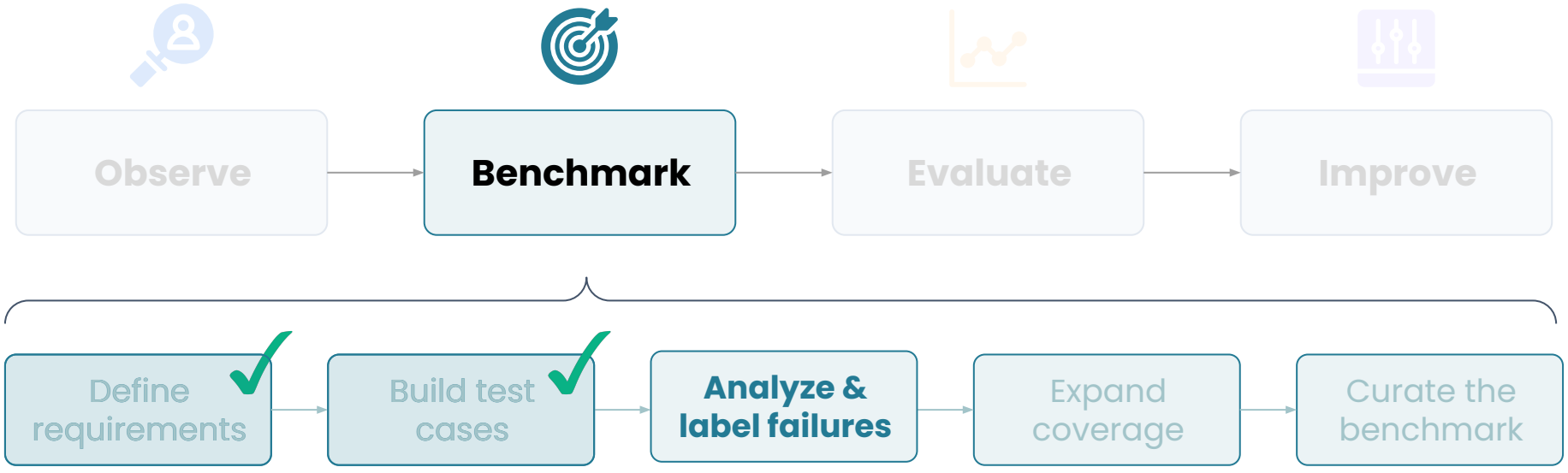


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Analyze and label failures

Module 2 · Video 3

Next step: Analyze and label failures



Manual review: label failures, group them and prioritize

Start by manually reviewing the failure

Requirement

Only match volunteers with dogs they can safely handle.

Test case



Experienced



130 lb, too strong

Expected outcome

Do not match

Agent response

Matched

Trace



What went wrong

UNRESOLVED

Human review



PASS



FAIL

Keep the information information from each run

For each run, write down:

- Outcome and explanation
- \+
- Trace
- \+
- Pending information

Test case



Experienced



130 lb, too strong

What went wrong

Matched a dog the volunteer couldn't safely handle

If you have an explanation you haven't checked, write it down as a possibility

Group cases by problem category

	Failure 1		Failure 2	
Test case	 Experienced	 130 lb, too strong	 First-time volunteer	 Reactive, needs experienced handler
What went wrong	Approved an unsafe match with the right information		Approved an unsafe match with the right information	

Group them together

Group cases by problem category

	Failure 1	Failure 2
Test case	 Experienced  130 lb, too strong	 First-time volunteer  Reactive, needs experienced handler
What went wrong	Approved an unsafe match with the right information	**Retrieved the wrong dog's record**
	Reasoning went wrong	**Information retrieval went wrong**

Use categories to label recurring failure patterns

Failure category	Description	Example
Judgment	The agent made the wrong decision	Unsafe volunteer-dog match
Groundedness	Response is not grounded in evidence	Retrieved the wrong dog's record
Execution	Did not carry out actions as required	Skipped eligibility check
Completion	Did not produce the intended result	Failed to effectively schedule an outing
Propagation	An error carried forward into later steps	Wrong record leads to wrong match

Assign the label to the failed case

Test case	 Experienced	 130 lb, too strong	 First-time volunteer	 Reactive, needs experienced handler
Broad error category	Judgment		Groundedness	Propagation
What went wrong	Ignored physical handling ability		Retrieved the wrong dog's record	

Update labels as you go

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Look at how often each problem showed up

Judgment

Reasoning error x 4

Scope violation x1

Groundedness

Hallucination x2

Faithfulness drift x3

Execution

Missing action x1

Completion

0 failures

Needs checking

Make sure to count problems in runs with more than one error

Harm depends on consequences

Dog shelter



Experienced



130 lb, too strong

Volunteer could lose control of the dog

High severity

Northwind

No return policy

I need to return these shoes

Sure. Returns are accepted within 30 days

Customer trust is lost

Moderate severity

Prioritizing what to fix

Failure type	Expected frequency	Severity	Priority
Inconsistent formatting	Common	Low: information still correct	Improve later
Unsafe match	Occasional	High: volunteer could lose control of the dog	**Fix now**



Experienced



130 lb, too strong



First-time volunteer



Reactive, needs experienced handler

Try similar situations



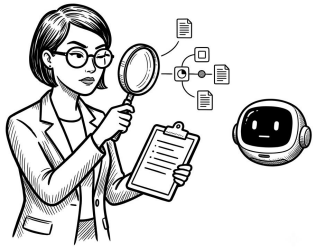
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Build a test set you can trust

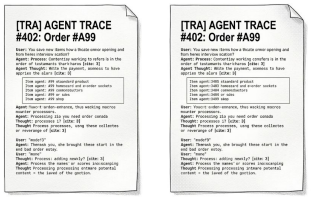
Module 2 · Video 4

A four-step workflow to build coverage

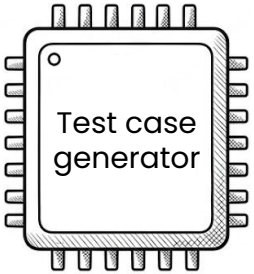
Expand coverage



Find coverage gaps



Add real examples



Generate missing cases



Curate the test set

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Look for clues in what you already have

Mistakes you found



Experienced volunteer



130 lb, too strong



Experienced, but not eligible for outings

Your requirements



First-time volunteer



45 lb, calm

Change one detail

Try a more difficult case

Real requests & bug reports





Examples you left out

New ones since then

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Fill coverage gaps with real examples

Experienced volunteer, not eligible for outings

Where to look:

- Manual review
- User requests
- Bug reports
- Support tickets
- Recorded runs

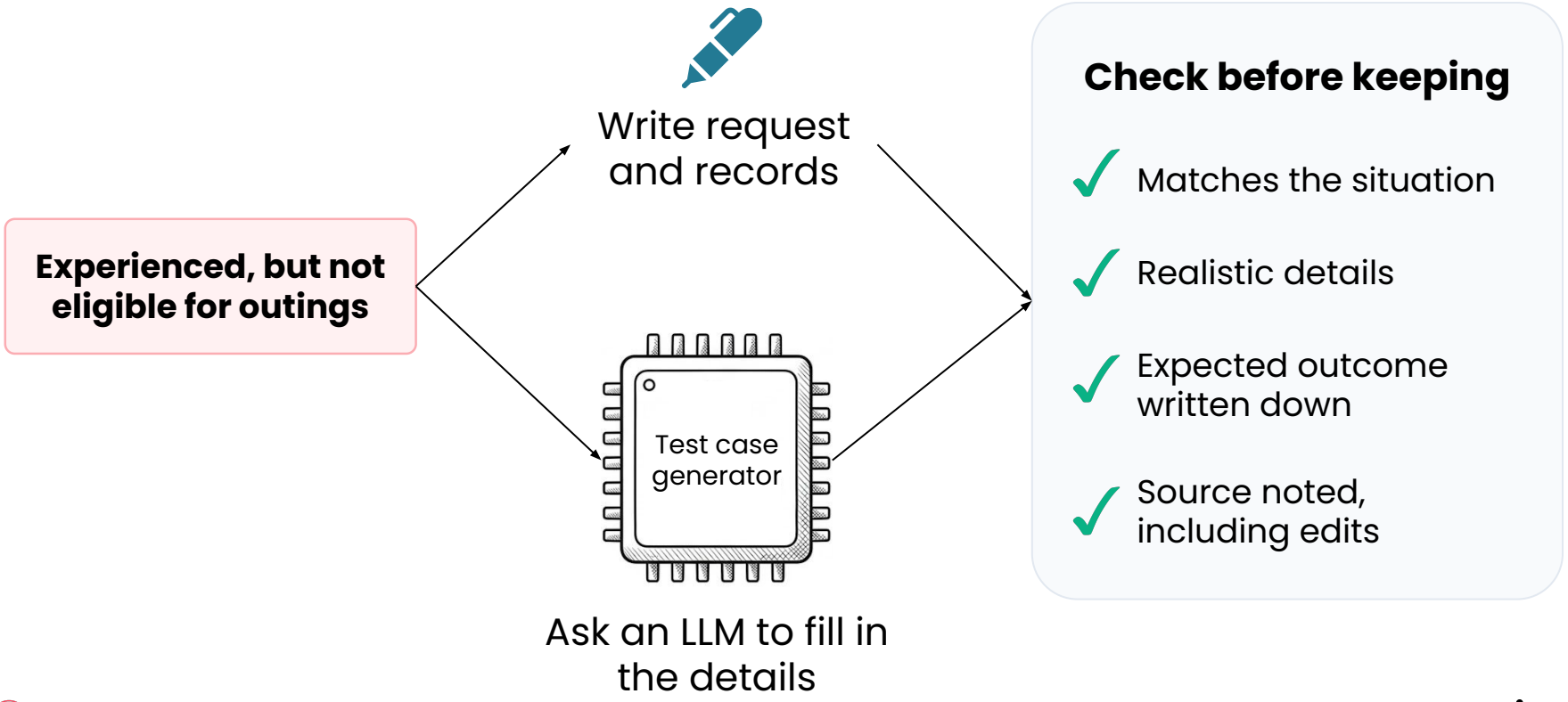


Support ticket: experienced volunteer, not eligible for outings

New test case

- Original request
- Volunteer record

Use synthetic data for gaps you still can't fill



Keep cases that test a real difference



Volunteer experience missing



Same case, different name

Remove: it repeats a check you already have



Dog handling notes missing

Keep: it checks something different

Same requirement, same expected outcome, still a different check

Curate the benchmark



Still needed?

Covers a meaningful situation	Mistake fixed →	**Keep:** it could come back
Protects against an important regression	Rule dropped →	**Remove**
Clarifies a requirement boundary	Rule changed →	**Update** the expected outcome

Include enough difficult cases in your benchmark

Test set A



98%

Straightforward: 98/98 passed

Difficult: 0/2 passed

Test set B

More difficult cases than in everyday use



The score describes your benchmark, not production metrics

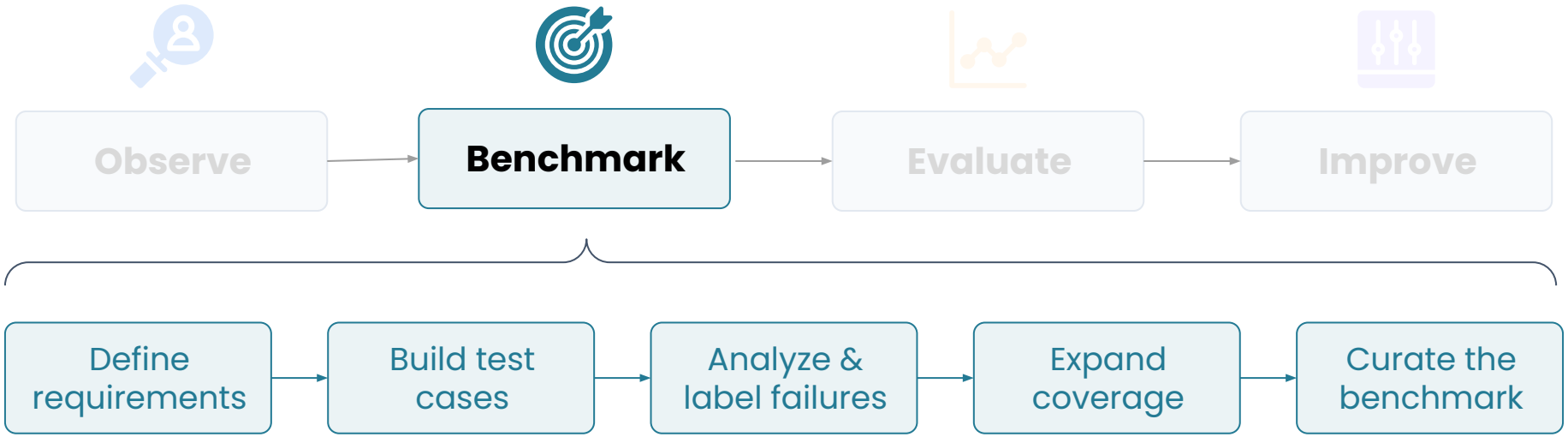


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Build the Northwind benchmark

Module 2 · Video 5

The benchmark workflow



Start with a behavior you already found

How should I care for my merino wool sweater?

Catalog:
`Machine washable on wool cycle`

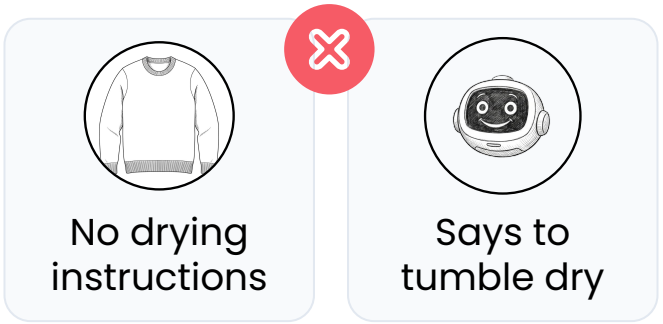


You can machine wash it on a wool cycle and then **tumble dry**



The agent recommended tumble drying, but the sweater didn't have drying instructions

Turn one observation into a broader requirement



Don’t recommend tumble-drying this sweater

Too narrow

Only make claims about products that are supported by Northwind data

Broader requirement

Does Northwind data support the claims in this response?

Review question

Save the expected outcome for each case

Situation	Customer query	Northwind data	Expected outcome
Complete info	What colors does the Classic Tee come in?	White, black, navy, heather gray	Answer with the listed colors; add none
Partial info	How should I care for my merino wool sweater?	Machine washable on wool cycle	Use the supported washing instruction; no drying advice
Missing info	What's your return policy?	No return policy information	Say the information is not available

Vary the customer side of the cases

What colors does the Classic Tee come in?



Do you have the Classic Tee in **size** medium and what **colors** are available?



→ **Keep both**

What colors does the Classic Tee come in?

Do you have the Classic tee in different colors?

→ **Keep one of them**

Compare each run with its expected outcome

Test case

What colors does the Classic Tee come in?

Catalog:
White, black, navy, heather gray

Expected outcome

Answer with the listed colors; add none

Agent response

Pass run

White, black, navy, heather gray

Fail run

White, black, navy, heather gray, and forest green

What went wrong

Named a color not supported by Northwind data

Failure label

Groundedness → Hallucination

Label why a case failed

Test case	What's your return policy?	**Northwind data:** No return-policy information
Expected outcome	Say the information is not available	
	Pass run	**Fail run**
Agent response	I do not have return Policy information available	You can return items within 30 days
Failure label		Scope violation

Look across the set to find coverage gaps

Questions to ask
Several unsupported product claims?
Invented policies outside of scope?
Cases where the agent should say information is missing?
Cases where the agent should answer directly?

Cases by topic

Care instructions ×6	Colors ×1
Materials 0	Sizing 0
Availability 0	Compatibility 0

Cases by expected response

Info is missing ×5	Answer directly ×1

Catch failures and confirm normal cases still work

Fill coverage gaps with real examples

Colors, materials, and sizing questions

Where to look:

- Traces
- Reviewed runs
- User questions
- Bug reports
- Support examples

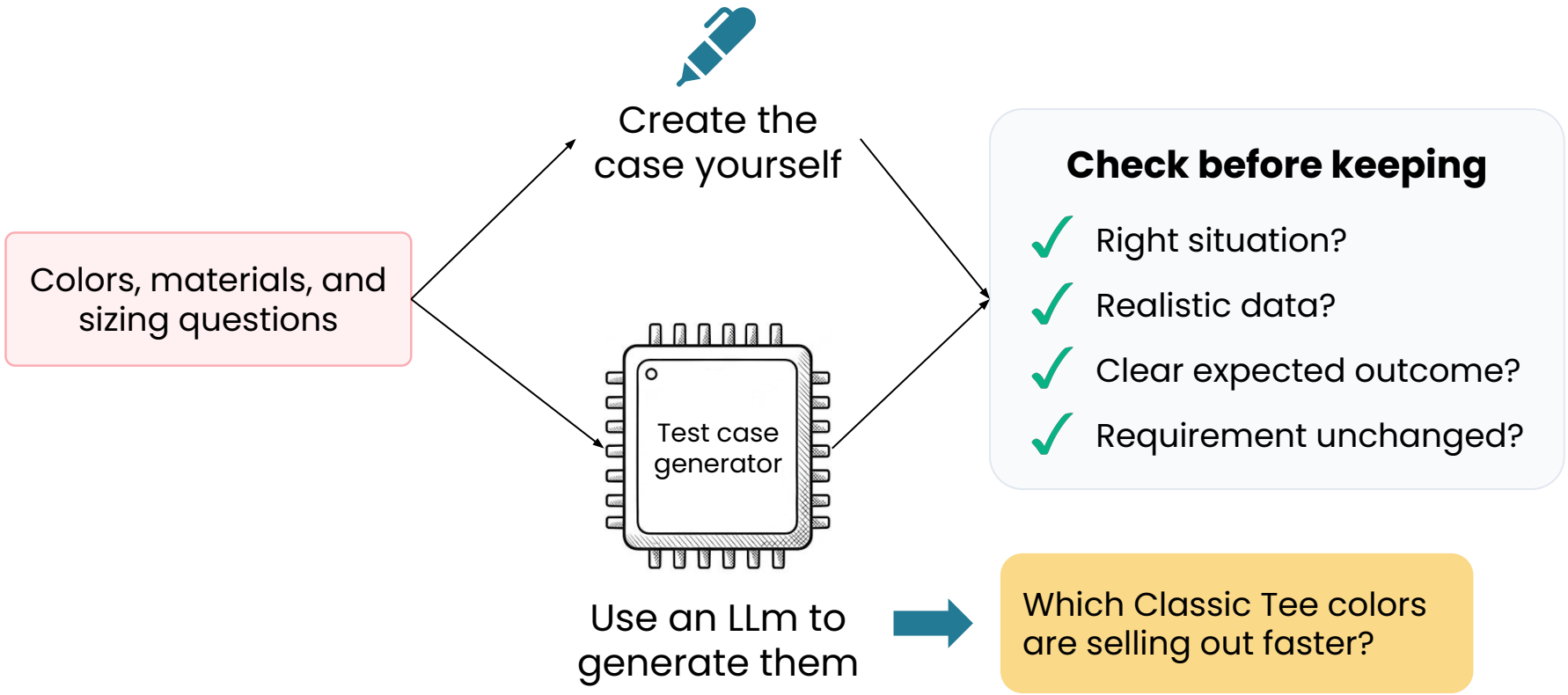


Real question:
"Can the sweater shrink in the wash?"

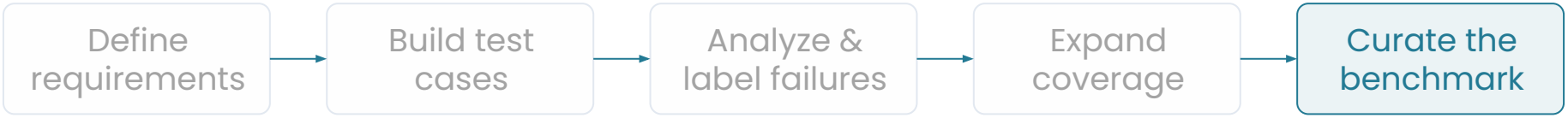
Add to benchmark

- Customer query
- Expected outcome

Use synthetic data for gaps you still can't fill



Curate the Northwind test set



Is the Classic Tee available in medium?

Do you have the Classic Tee in size M?

Remove one

Is the Classic Tee available in medium, and what is your return policy?

Keep

A useful case:

- Covers a meaningful situation
- Protects against an important regression
- Clarifies a requirement boundary

Include enough difficult cases to measure them

Test set A



98%

Straightforward: 98/98 passed

Missing information: 0/2 passed

Test set B

More difficult cases during development



Read it in context: benchmark ≠ production

What a complete benchmark case looks like

Customer query	Northwind data	Expected outcome	Source	Failure label
What's your return policy?	No return policy information	Say information is not available	Support example	Scope violation
How should I care for the Merino Wool Sweater?	Machine washable on wool cycle	Use only supported information	Trace	Faithfulness drift
What colors does the Classic Tee come in?	White, black, navy, heather gray	Answer with the listed colors	Synthetic	Hallucination

Your curated test set becomes a benchmark

1. Requirements
2. Test cases
3. Expected outcomes
4. Reviewed judgments
5. Failure labels

Northwind benchmark

Run it again as the app changes



Version 1

Compare versions against a stable standard



Version 2